

Plasma Spray Processors

Title: Operation

Ver 00 / 01.04.2019 QMS-11

ISO Clause: 8.4,8.5,8.5.1,8.5.2,8.5.3,8.5.4,8.5.5,8.5.6,8.6,8.7,7.1.3, 7.1.4, 7.1.4, 8.1, 8.5, 8.6, 8.7

01. **Purpose:**
To define methodology followed by Plasma Spray Processors for repair & service realization
02. **Scope:**
Produce and deliver equipment as per customer requirements
03. **Process Owner:**
Works Manager & DH Production
04. **Process Description:**
Customer requirements are satisfied through continuous interaction in all stages of Operation and till delivery of product.
05. **Key Process Indicators:**
01. On time delivery of product & service
02. Number of corrections required during and post Operation
03. Customer complaints
06. **Abbreviations:**

ISO International Organization for Standardization.
PFD Process Flow Diagram
CAPA Corrective & Preventive action
07. **Records:**

Sl. No.	Record Title	Format No.	Retention Period	Retention Resp.	Indexing Method
1	Production Planning	QF/MF/01	6 Months	HEAD Operation	Format
2	Machine Overhaul Record	QF/MAINT/08	Till Rev.	HEAD Operation	Computer/Hard copy
3	List of Machines & equipments	QF/MAINT/01	Till Rev.	HEAD MAINT	Computer / Format
4	Machine Brakdown Report	QF/MAINT/045	1 Year	HEAD MAINT	Computer/Hard copy
5	Preventive Maintenance Record	QF/MAINT/03	1 Year	HEAD MAINT	Computer/Hard copy
6	Job card / In-process Inspection report	QF/QA/06	Till Rev.	DH QA	Computer/Hard copy
7	Final Insp. Report	QF/QA/11	one year	HEAD QA	Hard copy
8	Master list of measuring & Test equipment	QF/QA/02	Updated every year	HEAD QA	Computer / Hard copy
9	Instruments & Gauge history card	QF/QA/03	Updated every year	HEAD QA	Computer/Hard copy
10	Material Inward Record	QF/ST/03	1 year	HEAD QA	Computer / Hard copy
11	Order Cum Inward Register	QF/PUR/04	1 year	HEAD QA	Hard copy
12	Customer Enquiry Monitoring for Repair	QF/MKT/01	1 year	HEAD Operation	Computer / Format
13	Tags	QF/MF/03		HEAD Operation	NA
14	Belt Stock Monitoring Sheet	QF/ST/01	1 year	HEAD STORE	Computer / Format
15	Process & Engineering change control sheet	QF/QA/09	Till Rev.	HEAD QA	Hard copy/ Computer

Disposition: Shred / Delete

Prepared by		Reviewed by		Approved by	
Designation	DH Operation	Designation	Partner	Designation	Partner
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Planning and Provision of Services

1. Purpose

- To ensure that required service / repair activities are carried out so as to meet requirements.

2. Scope

- This procedure is applicable for services / repair of products in workshop .

3. Responsibility

- Works Manager with the help of H Production & DH QA is responsible for implementation of this procedure.

4. Input

- Equipments & Products received for processing & repair.

5 Procedure

5.1 For Repairs of parts as per customer specifications

- Preliminary checks of existing specifications (Pre-inspection)
- cleaning of components is done .
- Components condition is verified as per Pre-inspection report.
- Wherever repair activities are required, necessary activities are carried out as per Job card & inspection report
- Work instruction referred in carrying of all activities.
- Inspection / test remarks based on repair activities are maintained in QF/QA/05 (Assessment Report) and QF/QA/06 (Final report)

5.2 Product non-conformities observed during repair / processing activities are recorded in nc record sheet.

5.3 Repair and service activities are planned so as to meet customer's delivery schedule. It is ensured that only authorized persons as carry out repair/service activities

6 Output

- Successful completion of repair / service activities is output of this process.

7 Effectiveness Monitoring

Effectiveness of this process is monitored through delivery performance on monthly basis and technical customer complaints on quarterly basis

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operational And Service Provision

1. Purpose

- To ensure that production & service provisions and parts supply are carried out under controlled conditions.

2. Scope

- This system is applicable to service activities, repairs, sales product identification, customer property preservation and dispatch.

Responsibility

Sales Manager & Works Manager for:

- Control of production process, product status and identification.
- Customer property preservation.
- Stores items accounting, preservation and traceability where required and dispatch of repaired Product.
- Partner for sales activity.

System (Service Provision)

4.1 Planning

- Processing & repairing is carried out as per customer requirements. Normal Servicing activities are cleaning, assessment, Plasma, HVOF, Arc, PTA and Flame Spray.

Please refer ANNEX-B1 for flow chart of service & repairs activity from enquiry receipt to Invoicing

4.2 Control over services (Cl. No. 7.5.1)

- Required scope of work / specifications / work instructions are made available for personnel in workshop / for processing.
- Repair activities are carried out under controlled conditions as per relevant work instructions.
- Processing / Repairs are carried out with appropriate tools / accessories, machinery and equipment. Performance of repair activity is monitored with help of suitable monitoring and measuring equipments and appropriate supervision.

Validation of Process for Service Provision (Cl.No.8.5.1f)

- The manufacturing activities are monitored and measured at appropriate operation stages.
- Dispatch of fully serviced and ready finished product is carried out after quality inspection as per customer requirements as defined.

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Property belonging to customers or external providers, Product Preservation and Dispatch

1. Purpose

- To control receipt, storage, maintenance and reporting of Customer Supplied Products (CSP) left in the care of Plasma Spray Processors to control receipt, storage and issue of spares

2. Scope

This procedure applies to all material and consumables purchased, intended to be incorporated into the repaired products. Component received for servicing/ repairing are also controlled as per this procedure

2. Responsibility

- Works Manager with the help of Stores & production is responsible for implementation of this procedure

3. Inputs

- Component received for processing, repairs. Bought out components
- ## 4. Procedure

5.1 Purchase items are verified with respect to P.O. and are recorded in QF/ST-01 & QF/ST-02. These items are then stored to designate storage areas. Stocks records of critical spares are maintained in QF-ST-01. Issues of spares are authorized by Works manager or Person authorized by Partner.

5.2 Receipt

a) All incoming CSP items are checked against customer order for quantity, description and visible damage before their particulars recorded in the incoming Inward register QF/ST/02 & incoming insp. report QF/QA/07.

b) The CSP shall then be stored for safe storage with the

5. Maintenance of CSP:

- CSP shall be appropriately identified, stored and maintained (e.g. applying antirust oil / grease etc.)
- In the event the product is damaged in transit or becomes unsuitable for use, the Store keeper shall record the damage in Inward Register. The Partner shall then inform the customer. Necessary actions in such cases shall be decided in consultation with customer.

6. Packaging of Repaired CSP

- Generally repaired components are packed with polythene bags & corugated box..
- Wherever original packing is not found to be suitable appropriate packing box is prepared. This shall be done in consultation with customer.

7. Dispatch / Transportation of components shall be carried out as below:

- Components are dispatched through customer specified transporters.
- Wherever customer has not specified any transporters, dispatch is arranged in such a way that component will directly reaching to customer site without any change in transport vehicle in between.

7. Output

- Proper preservation of customer supplied product till dispatch

8. Effectiveness Monitoring

- It is monitored on quarterly basis through customer complaints related to packaging / product preservation.

9. Cross Reference

- | | | |
|-----|---|-----------|
| 8.1 | External provider Rating Report /Assessment | QF/PUR/02 |
| 8.2 | Production planning | QF/MF/01 |
| 8.3 | Belt Stock Monitoring Sheet | QF/ST/01 |

Note: We are not keeping external provider products

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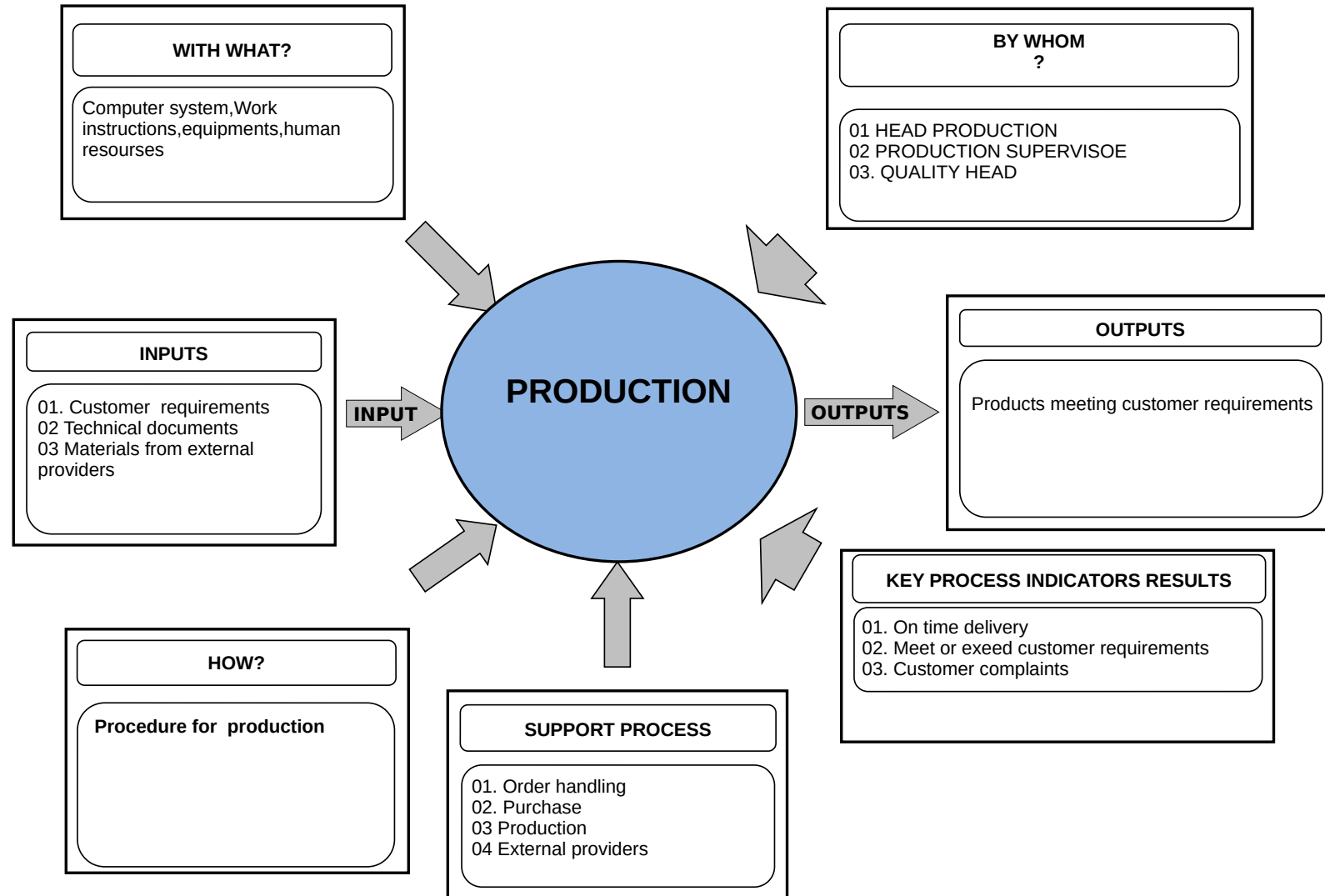
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Source	Input	Description of Activity in Flow Chart	Out put	Risk considered	Control methods
Customer requirements on new products or /modification of current products & equipments	Customer order Competancy levels Organisational capability	Repair & Service planning	Order wise production & repair plan	Over commitment	CFT review
	Order wise plan /Daily plan	Organise resources like materials, machinery, manpower and external providers and detail the production process	New work instructions or updating existing checklist & WI Verification of testig & repairing procedures	Vague /incomplete instruction	CFT review
	Work instructions Customer Drawings Work procedures	Repair & service as per PreInsp. report, Job Card, WI & Checklist	WIP equipment	Errors in processing & teting	Corrective action
	Repair & Testing Procedure	Test product as per specification/ Job card/ Checklist & Drawings	Tested equipment	Misunderstanding of customer requirements	Written communication from customer
	Tested product, Packing instructions	Identification/ leveling of tested products at in-house & verification sticker repaired at customer end		Mismatch between packing, ID instructions and product	Process monitoring
	Packed / Finished Equipments	Despatch/ Handover	Despatch / handover documents	Wrong taxation or address or improper documents	Close review during invocing to avoid such mishaps
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ISO Clause: 8.5.1 (f)- Process Validation

Source	Input	Description of Activity in Flow Chart		Out put	Risk considered	Control methods
Drawings, Man, Machine, Method, Materials. Plasma, HVOF, Arc, PTA and Flame Spray re identified in-house for process validation at In-House.	Customer Drawings	Review welding & coating requirements required / or confirm with cust.		Specification File	Lack of awareness of Process validation	CFT review
	Competancy levels of Operator	Validate competency level of Operator		Competency Matrix Skill Matrix	Inadequate competency validation	Criteriata defined in competency matrix
	Welding Machines	Ensure maintenance status/ calibration status of Processing equipments		PM Record Calibration Record	PM not done as per plan Calibration not done as per plan	PM & caln. Plan made Review in MRM
	Welded product	Produce sample piece		Processed Piece	Wrong Processing	Competency validated Operator training
	Tested product	Conduct in-hose destructive test or send it to outside lab for testing		Test Certificate	Wrong testing	Testing mechanisn defined
	Result	Record validation record		Process Validation record Revalidation record	Wrong documentetion	Format made for process validation & revalidation
Prepared by		Reviewed by		Approved by	partner	
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ISO Clause: 8.5.2, 8.5.3- Identification & Traceability/ Property belonging to customers or external providers

ISO Clause: 8.5.4- Preservation

Source	Input	Description of Activity in Flow Chart		Out put	Risk considered	Control methods	
Customer supplied Drawings, Incoming materials, WIP Materials, Finished products	Customer Property/ Drawings External provider property	Make proper entry of customer supplied equipments received for repair. Make list of drawings, if supplied by customer ↓		Drawings File Inward record	Not identified CSP	Awareness training provided Formats made in QMS	
	Incoming materials	Identified Incoming materials by tags with details, Part desc., status, qty. ↓		Identification Tags	Not identified by Tags	No materials kept without having ID Tags	
	WIP Parts	On stock items, only component type and specification are put. On customer items requiring further repair, a treceability number is generated and maintained. On complete assembly only the component type and customer name will be mentioned ↓					
	Finished parts	Identification of in process and finished component shall be done by tag attached with particular component in receiving stage ↓			Not identified by Tags	No materials kept without having ID Tags	
	Traceability of finished parts	Components traceability shall be maintained up to dispatch by means of tags which contains type of Component / Customer name and Plasma SprayTraceability no.During component inspection, Plasma Spray traceability no. is referred in relevant inspection reports			ID label not puts on	Verified before dispatch	
					finished equipments	Trained to operators	
	Incoming/WIP & Finished products	Preserve products by keepinn incoming in packing bags/ Racks & Bins. Keep WIP & repaired equipments duly applying rust preventive oil			Proper paking & proper keeping	In-proper storage	Provision made for proper storage of WIP, Receipt & FP
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